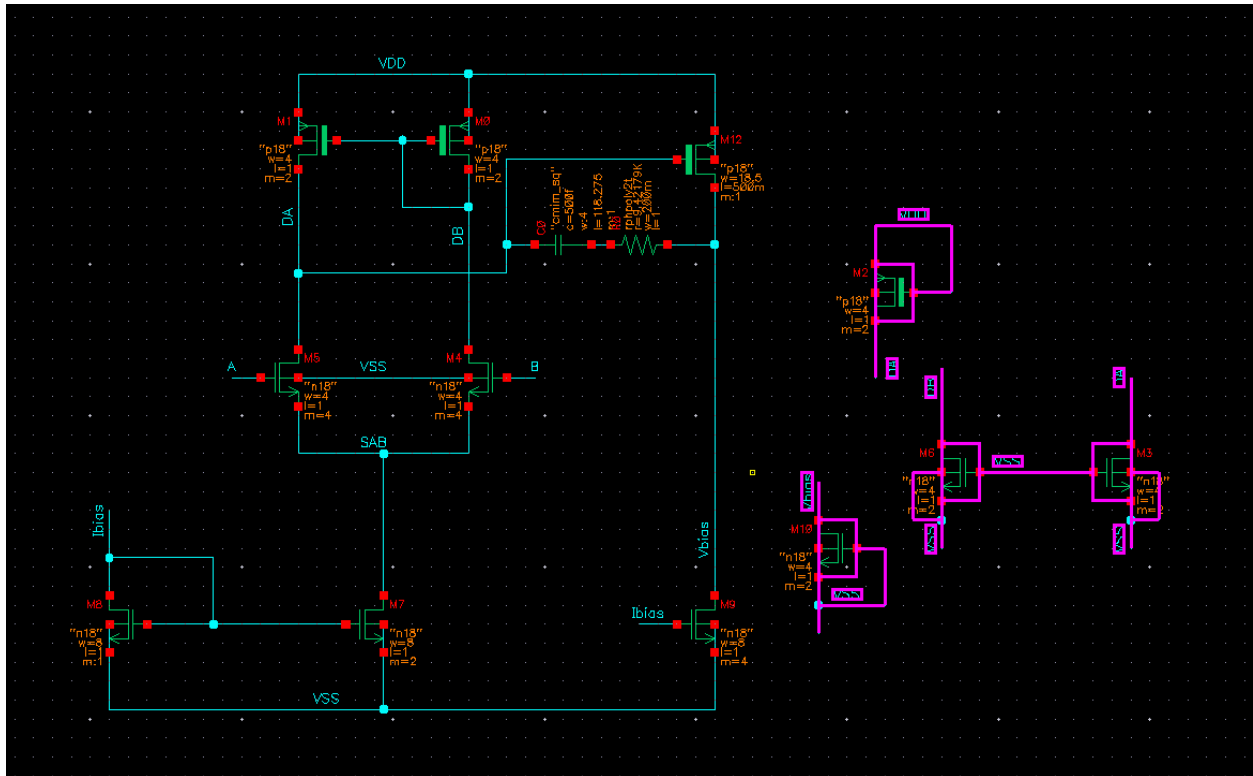


Two stage opamp Schematic:-



LVS report:-

Cell Final_OPAMP_2_Stage Summary (Clean)

CELL COMPARISON RESULTS { TOP LEVEL }

```

#          #          #####
#          #          #          #
#          #          #          #
#          #          #          #
#          #          #          #
#          #          #####

```

```

-   -
*   *
|
\___/

```

Warning: Unbalanced smashed mosfets were matched.

LAYOUT CELL NAME: Final_OPAMP_2_Stage
SOURCE CELL NAME: Final_OPAMP_2_Stage

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Nets:	9	9	
Instances:	28	21	* MN (4 pins)
	7	7	MP (4 pins)
Total Inst:	35	28	

NUMBERS OF OBJECTS AFTER TRANSFORMATION

	Layout	Source	Component Type
Nets:	9	9	
Instances:	8	8	MN (4 pins)
	4	4	MP (4 pins)
Total Inst:	12	12	

* = Number of objects in layout different from number in source.

Cell Final_OPAMP_2_Stage Summary (Clean)

***** INFORMATION AND WARNINGS *****

	Matched Layout	Matched Source	Unmatched Layout	Unmatched Source	Component Type
	-----	-----	-----	-----	-----
Nets:	9	9	0	0	
Instances:	8	8	0	0	MN(N)
	4	4	0	0	MP(P)
	-----	-----	-----	-----	
Total Inst:	12	12	0	0	

o Statistics:

34 layout mos transistors were reduced to 11.
 23 mos transistors were deleted by parallel reduction.
 26 source mos transistors were reduced to 10.
 16 mos transistors were deleted by parallel reduction.

o Matched Mosfets Which Have Been Unequally Reduced:

X61/X0/M0 {8.230, 3.925}	MM7
X63/X0/M1 {22.090, 3.925}	MM7==2
X63/X0/M0 {20.550, 3.925}	** missing smashe
X61/X0/M1 {9.770, 3.925}	** missing smashe
M0 {5.150, 3.925}	MM9
X63/X1/M1 {25.170, 3.925}	MM9==4
X63/X1/M0 {23.630, 3.925}	MM9==3
X62/X1/M1 {19.010, 3.925}	MM9==2
X62/X1/M0 {17.470, 3.925}	** missing smashe
X61/X1/M1 {12.850, 3.925}	** missing smashe
X61/X1/M0 {11.310, 3.925}	** missing smashe
M1 {6.690, 3.925}	** missing smashe

DRC Check:-

Only fabrication errors are there, rest all are clear.

File View Highlight Tools Window Setup Help

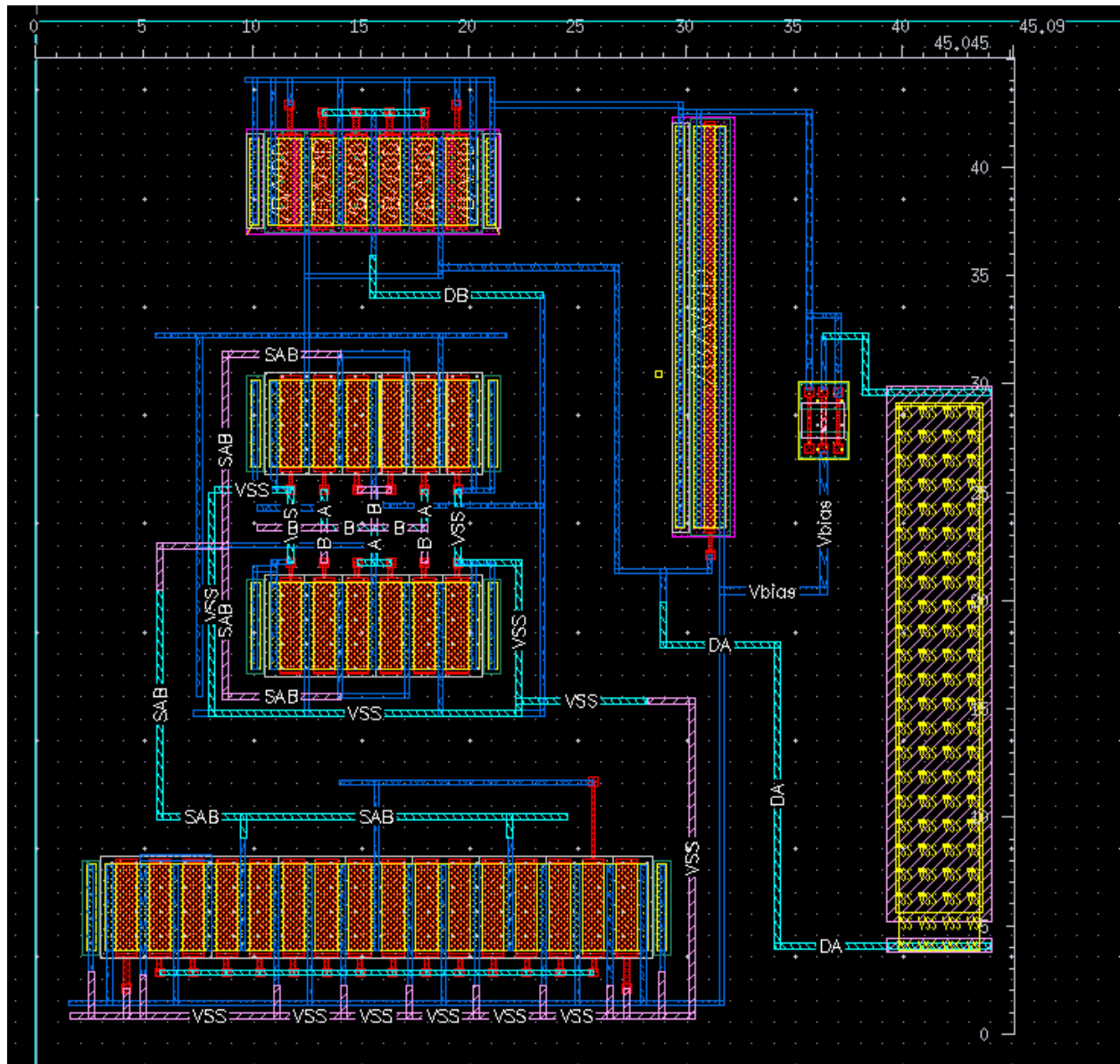
Filter: Show All Two_stage_opamp, 1123 Results (in 11 of 12 Checks)

Check / Cell	Results	Flat
Check 90	28	28
Check 11D	17	17
Check 13D	9	8
Check DC1	64	127
Check DC2	1000	1024
Check 3R_LT15	1	1
Check 6W	1	1
Check 9X	1	1
Check 11X	1	1
Check 19X	1	1
Check 13X	1	1
Check DENSITY_PRINT_FILES	0	0

Rule File Pathname: _DRC.header_

90: M1 area must be min 0.202 sq.um (90).

Layout:-



Area = 2025 μm^2

Pex Report:-

File Settings Configurations Help

Custom Rules Inputs Outputs LVS Run Control Search Transcript

merged nets = 0
error nets = 0

=====

CALIBRE xRC WARNING / ERROR Summary

xRC Warnings = 7
xRC Errors = 0

=====

--- CALIBRE xRC::FORMATTER COMPLETED - Thu Apr 23 01:00:05 2026
--- TOTAL CPU TIME = 0 REAL TIME = 0 LVHEAP = 4/11/336 MALLOC = 74/74/74 ELAPSED TIME = 1

WARNING:Hostname to
Checkout succeeded: ca
License file:
License Ser

WARNING: As of version
WARNING: As of version
WARNING: No ground ne

*** xRC run finished with exit code 0 ***

Run PEX

Start RVE

Calibre Info

Calibre View generation completed with 0 WARNINGS and 0 ERRORS. Please consult the CIW transcript for messages.

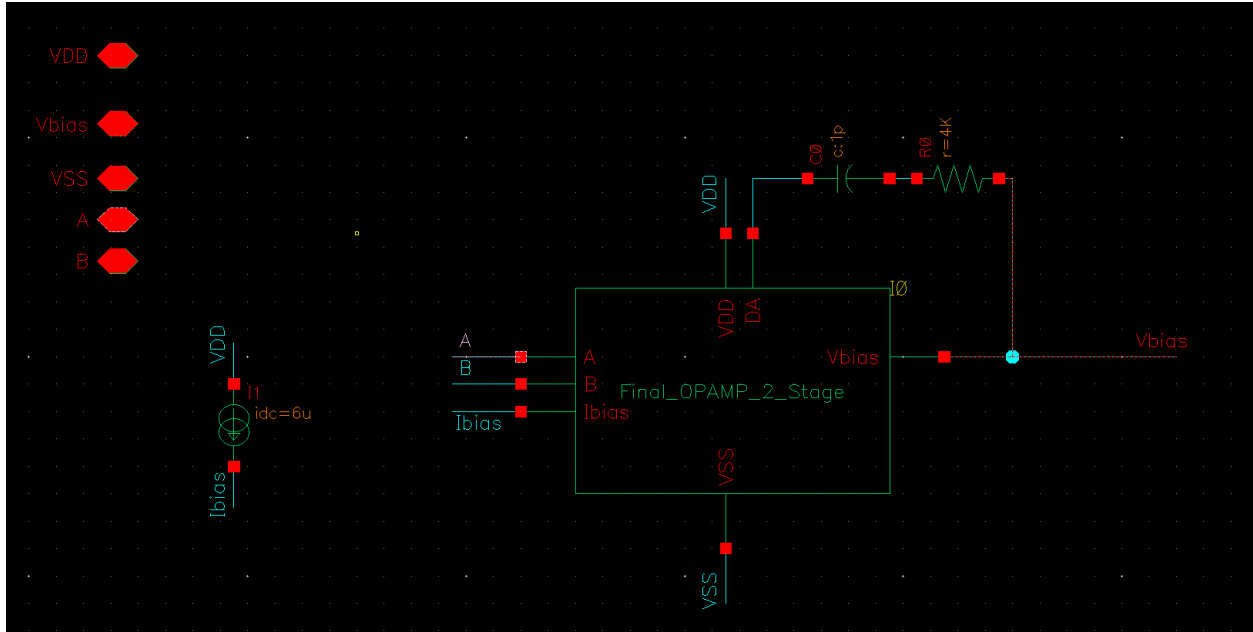
Close

0 Errors, 22 Warnings, 6 Infos

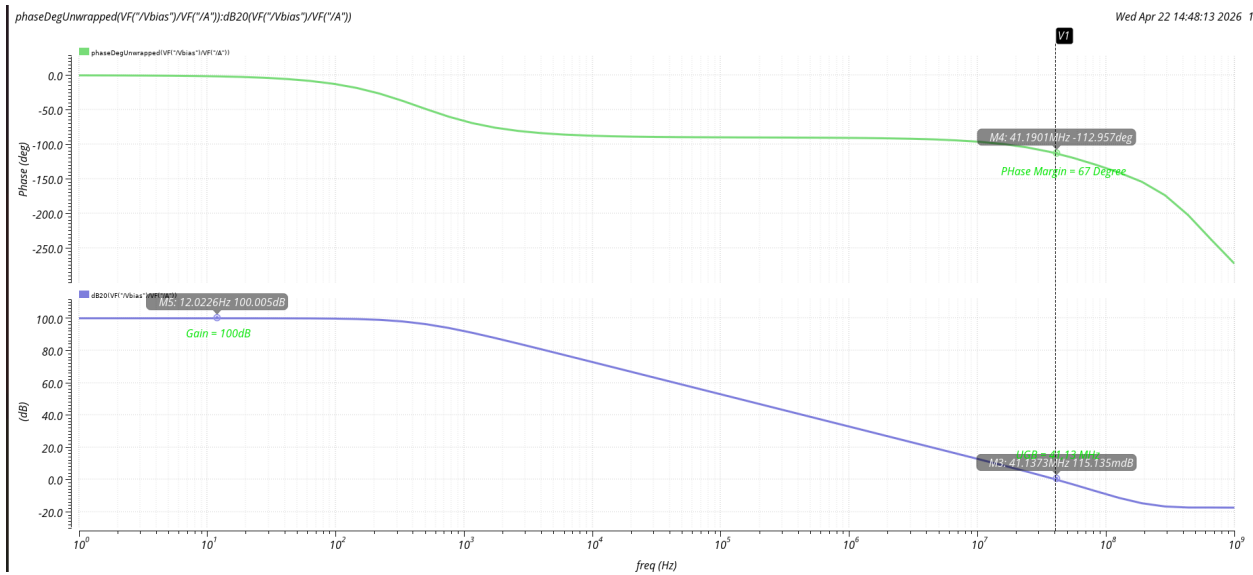
Line	Type	Description
1	Info	Verifying the source netlist is complete before starting the run
87	Warning	Please enable "throughput-performance" profile for best performance
88	Warning	Please increase max process limit (4096) to system limit (2063238)
89	Warning	Please increase descriptors limit for best performance (4096)
92	Info	// CPU Info: Cores = 40, SMT enabled with 40 additional virtual processors
93	Warning	Please enable "throughput-performance" profile for best performance



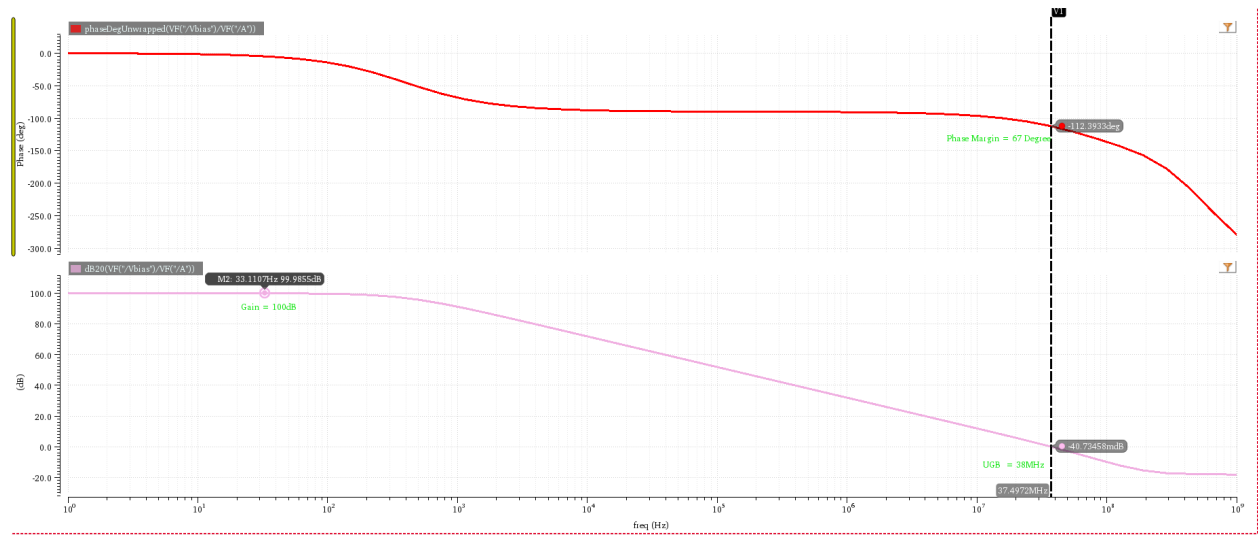
Test Bench Schematic:-



Pre Layout Simulation:-



Post layout simulation:-



Comparison Table:-

	<u>Gain</u>	<u>UGB</u>	<u>Phase margin</u>
<u>Pre layout</u>	99.98dB	41.37MHz	68.8 Degree
<u>Post Layout</u>	100.05dB	37.39MHz	68.3 Degree

Comparison:-

The post-layout results show a slight drop in UGB (from 41.37 MHz to 37.39 MHz), mainly due to parasitic capacitances and resistances introduced during layout, which slow down the circuit. The gain remains almost unchanged (~100 dB), indicating good robustness of the design against layout effects. The phase margin also shows a negligible reduction (68.8° to 68.3°), suggesting stability is largely preserved after layout.